裕太微电子
MotorComm

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YT8512H YT8512C APPLICATION NOTES

Single Port 10/100 Ethernet PHY Transceiver

V 2.3

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Revision History

Revision	Release Date	Summary
1.5	2020.05.27	Modified document format
1.6	2020.05.28	Corrected typos: "RGMII" was changed to "MII/REMII/RMII"; Corrected template test register 0x27 to 0x2027; Corrected the value of template10 Base-T test register 0x200a
1.7	2020.06.15	Corrected section 4.8.2.2: TXC of RMII1 should be an input, and added power-on reset sequence reminder
1.8	2020.07.03	Added introduction to sleep function
1.9	2020.08.03	Modified the description
2.0	2020.08.20	Added descriptions about phyaddr0 and broadcast address
2.1	2020.11.09	Corrected section 4.8.2.2: RMII externally input 50MHz application circuit connection and added more descriptions Added section 4.9: avoid entering SCAN mode during circuit design
2.2	2020.11.26	Corrected section 4.8.2.2: RMII externally input 50MHz application circuit connection and recommended input clock via XTAL_IN
2.3	2020.12.17	Corrected the description of section 6.4: LED configuration

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1. Brief description

YT8512 is a single-port, low-power 10M/100Mbps Ethernet PHY transceiver that supports both MII and RMII interfaces to connect to MAC. It mainly consists of power supply, reset, clock, data interface, LED output, etc., among which, the data interface includes xMII, SMI and MDI modules. This document mainly introduces some notes on hardware circuit and software-related configurations.

2. Abbreviations

YT: Motorcomm Electronic Technology Co., Ltd., refers to our company.

PHY: physical layer chip, used in this document refers to the Ethernet physical layer chip, i.e., YT8512 chip.

MAC: Media Access Control, data link layer, used in this document refers to the upper layer interface chip connected to PHY, usually a switch chip or CPU chip with integrated MAC function.

DUT: Device under test, used in this document generally refers to YT8512 chip.

LP: Link Partner, the pair-end connected chip, used in this document generally refers to the pair-end chip connected to the DUT through the network cable, fiber or SMA line.

UTP: un-twisted pair, the interface used to connect to other PHY chips, used in this document refers to the Ethernet electrical interface (or RJ45), including 100/10Mbps rates.

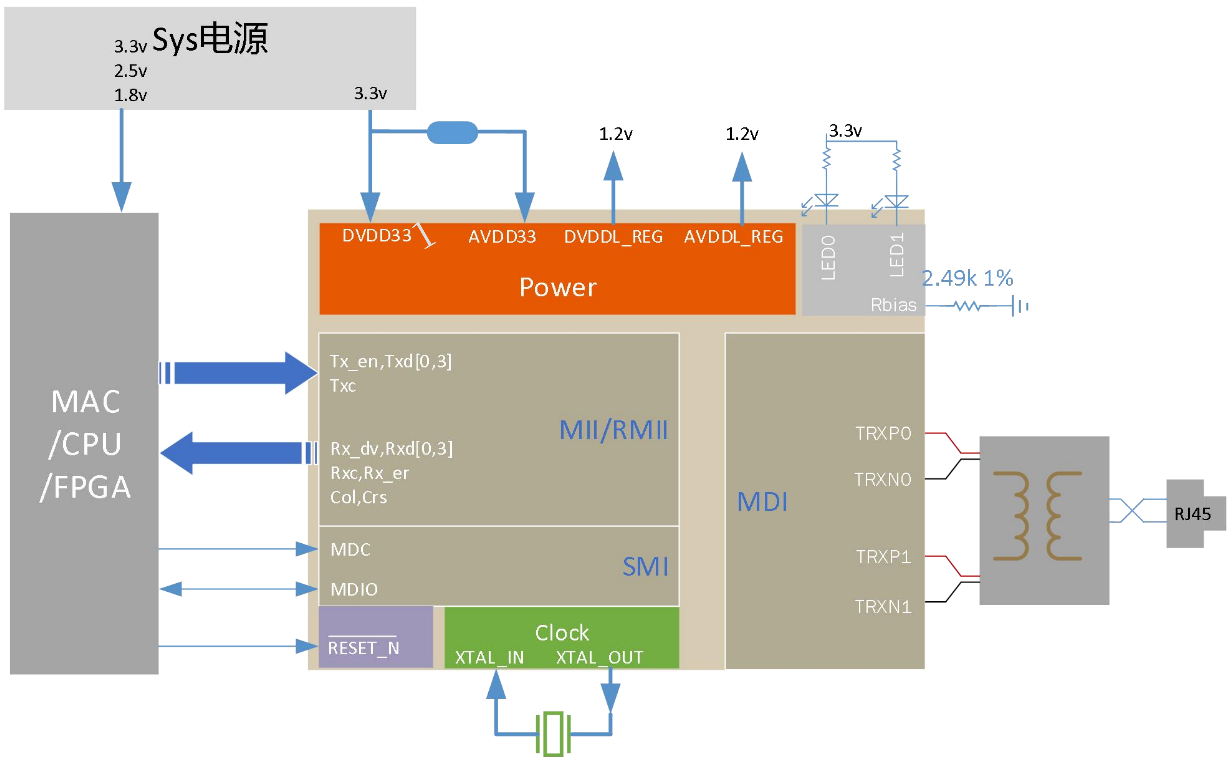
MDIO/SMI: Management Data Input Output, an interface composed of MDC/MDIO, is used to read and write the internal registers of PHY.

Mii: Media independent interface, the interface connecting the Ethernet PHY chip to MAC, only applies to 100/10Mbps rate. As the standard register of PHY specified in Section 802.3 is named MII register, the standard register is named Mii register in this document, which is recorded as mii_reg.

Ext: The number of MII registers is limited, which is 32 only. In order to meet the demand for more functions, the phy chip is configured with more registers, which is called extended register and recorded as ext_reg.

MMD: MDIO Manageable Device, refers to the register managed through the MDIO interface specified by IEEE802.3 standard Clause 45, which is called MMD register and recorded as mmd_reg.

3. Block diagram



4. Hardware circuit related

4.1 Power supply

The pins DVDD33 (pin14) and AVDD33 (pin7) are externally connected to 3.3v@80mA voltage, to supply power to the digital part and analog part of YT8512 respectively. It is better to connect a magnetic bead between the two pins, and connect 1uF and 0.1uF capacitors to each of the pins respectively.

DVDDL_REG (pin29) and AVDDL_REG (pin2) are the output pins generated internally by YT8512 that supply 1.2v voltage to the digital and analog parts respectively. When the two pins are in use, it is recommended to respectively connect external 1uF and 0.1uF capacitors close to the pins.

DVDDL_REG and AVDDL_REG pins are used to supply power to the internal circuits of YT8512 only, and cannot supply power to other devices externally.

4.2 Power-on sequence

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RESET_N (pin21) achieves hardware reset function, and the low voltage is active. When RESET_N starts to operate, the voltage is low. During this period, the power supply pin voltage and clock signal grow from scratch and keep stable, and then, RESET_N is released. Generally, the RESET_N can be reliably reset when it remains low for 10ms. The register values in YT8512 are restored to their default values after the hard reset. The specific sequence relationship is as follows:

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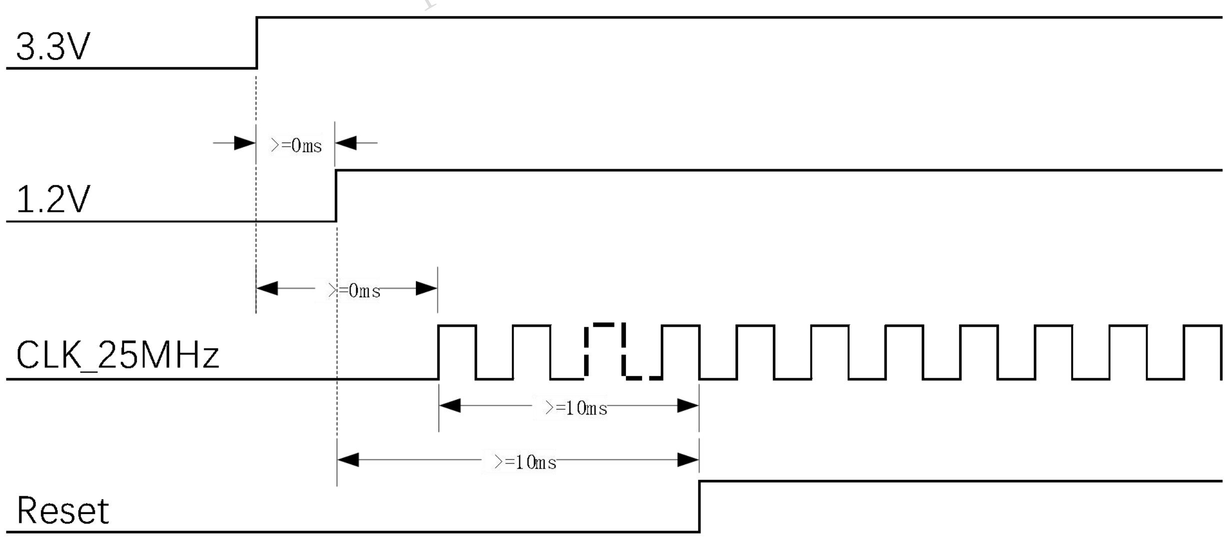

Notes.

1. If 1.2V or CLK_25MHz is generated by the chip's internal power supply/crystal oscillator circuit, the sequence between them and 3.3V is controlled internally by the chip to meet the chip's internal sequence requirements. It may be that 3.3V rises slowly while 1.2V rises fast, therefore, 1.2V stabilizes before 3.3V, which is a normal phenomenon and there is no problem. Or, it is normal that CLK_25MHz has output when the 3.3V power is not stable.

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2. The above sequence should be observed if 1.2V or CLK_25MHz is externally input. Among them, 1.2V and CLK_25MHz are not required to be sequential, but 1.2V and CLK_25MHz are required to be stable and held for at least 10ms before Reset signal is released.

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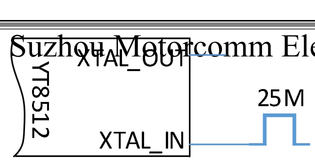
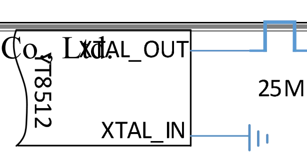
3. Generally, the ramp time of power supply must be at least 100us to prevent the chip from internal overvoltage.

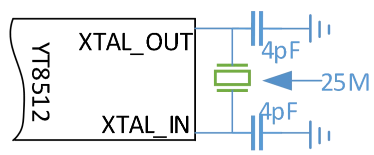
4.3 Clock

YT8512 can use 25M crystal oscillator, or external 3.3v 25MHz clock signal.





XTAL_IN connected to 25M clock source XTAL_OUT connected to 25M clock source



Using 25M crystal oscillator

YT8512 can also be connected to a 50MHz clock signal (replace the clock source or crystal oscillator in the above figure with 50MHz clock signal). If a 50MHz clock is used, the following registers need to be initialized after the system is powered on:

Write_ext_reg0x50[6]: 1’b1 // Bit6 of extension register 0x50 is set to 1 to select the clock frequency division factor

Write_mii_reg0x0[15]: 1’b1 // Bit15 of the basic register 0x00 is set to 1 to perform a soft reset operation

For details about how to connect the 50MHz clock, please refer to the “RMII1/2(50MHz clock)” section.

4.4 Rbias

The Rbias pin (pin1) requires an external 2.49k (1% accuracy) resistor connected to ground, and some of the internal electrical parameters of the YT8512 are related to this resistor.

4.5 POS

LED0, LED1, RX_DV, RXD1 and RXD3 are multiplexed pins. In addition to their functions in the normal state, at the initial power-on phase, the YT8512 detects the state of these 5 pins and makes some basic configuration of the chip according to the state. This process is called Power on strapping (POS).

These 5 pins have weak pull-down resistors inside the IC. If no external configuration is made, the default state is 0 in the POS phase. In practice, it is highly recommended to connect external pull-up or pull-down 4.7k resistors and not to rely on the internal weak pull-up/down resistors.

LED0(pin24)\LED1(pin25):

The states of LED0 and LED1 in the POS phase determine the PHY address of YT8512.

Pin name_phy address bit		phy address
Led1_add1	Led0_add0	
00		0
01		1
10		2
11		3

It should be noted that when the same MCU or MAC manages multiple phys, do not use 0 address. Since address 0 is a broadcast address, all phys will respond when the phy with address 0 is operated.

RXD1(pin10):

The state of RXD1 in the POS phase determines whether the LED0 pin is used as an LED or WOL output

RXD1-0: LED0 is used as LED output pin

RXD1—1: LED0 is used as WOL interrupt output (only sink current for open-drain mode)

RX_DV(pin8):

RX_DV - 0: MII mode

RX_DV—1: RMII mode

RXD3(pin12):

RXD3-0: TXC is the output

RXD3-1: TXC is the input

4.6 LED

YT8512 has two LED pin outputs, that's pin24-LED0 and pin25---LED1. Both pins have internal weak pull-down resistors.

In addition to being used as LED output pins, they are also used as phy address configuration pins during the power on strapping phase, so sometimes these two pins have a strong pull-up or pull-down (4.7k) externally.

The output polarity of the LED pin (that is, active high or active low) is related to the pull-up resistor or pull-down resistor connected to the pin. (If there is an external pull-up or pull-down resistor, the external one shall prevail; if there is no external pull-up or pull-down resistor, it shall rely on the internal default pull-up or pull-down resistor.)

If there is a pull-up resistor, it is active low (that is, it needs to be connected to the cathode of an external LED); if there is a pull-down resistor, it is active high (that is, it needs to be connected to the anode of an external LED)

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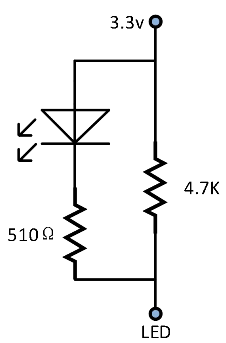

图4 a 上拉、sink模式

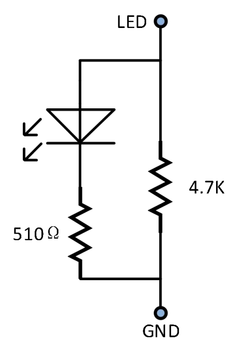

图4 b 下拉、source模式

Fig. 4a sink mode for pull-up resistor Fig. 4b source mode for pull-down resistor

As shown in the figures above:

1. During power strapping, LED0 and LED1 are pulled up, then in the working mode, LED is active low with sink current, the default LED behaviors are as follows:

   LED0: with 10M link up, the light is on and the output is low; with 10M data transmission, the light is flashing and the rest of the state lights are off.

   LED1: with 100M link up, the light is on and the output is low; with 100M data transmission, the light is flashing and the rest of the state lights are off.

2. During power strapping, LED0 and LED1 are pulled down (default pull-down inside the chip), then in the working mode, LED is active high with source current, the default LED behaviors are as follows:

   LED0: with 10M link up, the light is on and the output is high; with 10M data transmission, the light is flashing and the rest of the state lights are off.

   LED1: with 100M link up, the light is on and the output is high; with 100M data transmission, the light is flashing and the rest of the state lights are off.

The internal registers are required for changing the behavior of the LED. See the LED configuration section for details.

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4.7 SMI (MDIO) interface

The MDIO interface includes two signal lines: MDC (pin22) and MDIO (pin23). The MDC of YT8512 supports a data clock frequency up to 12.5MHz, and MDIO supports bi-directional data exchange. It should be noted that the MDC clock is not related to the TXC and RXC clocks.

Although there is an internal pull-up, when MDIO is in use, the MDIO needs to be externally pulled up to DVDD33 using a 4.7k resistor.

4.8 MII, REMII, RMII interfaces

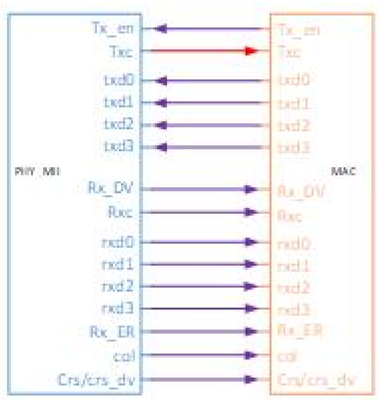
YT8512 supports MII/RMII interfaces to connect to MAC. YT8512 detects (Power Strapping) the RX_DV pin (pin8) during the initial power-on period. If RX_DV is low, it is set to MII; otherwise, it is set to RMII. In addition, because the RX_DV pin is a weak pull-down internally, if the external circuitry is not set, the YT8512 is the MII interface by default. However, it is strongly recommended for practical applications to add pull-up/pull-down resistor externally and not rely on the internal weak pull-up/down resistors.

YT8512 will also detect the pin RXD3/CLK_CTL (pin12, internal weak pull-down) during power-on, and determine the clock direction of TXC according to its level state. TXC is output at low level; TXC is input at high level.

The combination of these two POS produces the following pattern:

RXDV	RXD3	Mode	Remark	Application scenario
0	0	MII mode	TXC and RXCare output by PHY	Connected to MAC MII interface
0	1	REMI mode	TXC and RXC are output to PHY by the opposite end	Generally not connected to MAC, but to another PHY
1	0	RMII2	TXC is output from PHY to MAC	Connected to MAC RMII (TXC input) interface
1	1	RMII1	TXC is output by MAC to PHY	Connected to MAC RMII (TXC output) interface

4.8.1 MII and REMII modes



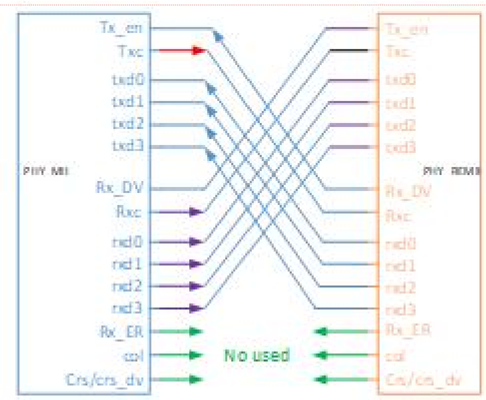


Figure 5a. YT8512 working in MII mode (Rx_DV and RXD3 are externally connected to 4.7K resistor to ground)

Figure 5b. YT8512 working in REMII mode (Rx_DV is externally connected to 4.7 K resistor to ground, and RXD3 is connected to 4.7K resistor to 3.3V)

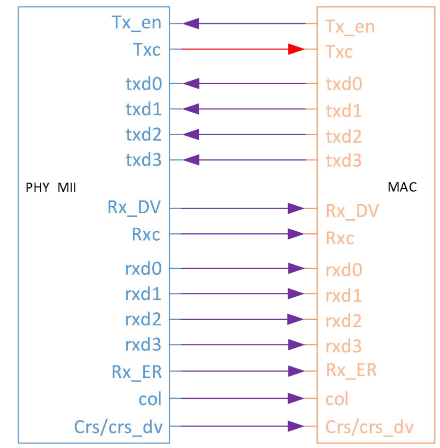


图5 a YT8512工作在MII状态
(Rx_DV、RXD3外接4.7k电阻到地)

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Figure 5a is a connection diagram of a typical Phy working in MII mode with MAC. Figure 5b is generally used in applications where two phys are connected back-to-back.

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4.8.2 RMII1 and RMII2 modes

The RMII modes of the PHY are divided into two types, RMII1 and RMII2. The difference between them is the direction of the TXC clock.

In RMII1 mode, TXC is the input.

In RMII2 mode, TXC is the output.

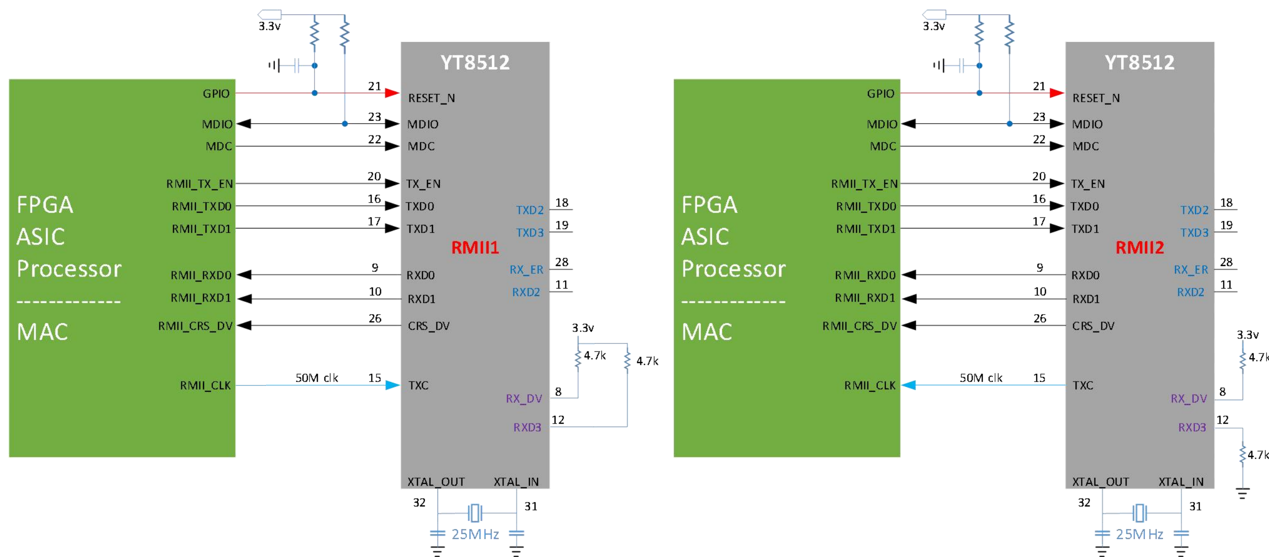
The RMII mode on the MAC side should correspond to the PHY mode, that is, PHY RMII1 + MAC RMII2, or PHY RMII2 + MAC RMII1.

Generally, because the RMII mode uses 50MHz RMII TXC, the customer may directly connect the RMII TXC sent by the MAC to the PHY as a reference clock to save the crystal of the PHY. Therefore, according to different clock sources, the circuit connection methods in RMII mode include: RMII1/2 (25MHz clock) and RMII1 (externally input 50MHz clock)

4.8.2.1 RMII1/2 (25MHz clock)

When 25MHz is used as the reference clock, it only needs to set the POS to the correct mode, and no software configuration is required.

The following figure shows the connection method between PHY and MAC by using a 25MHz crystal as a reference clock (the external 25MHz clock signal is similar, please refer to the previous section on clocks):



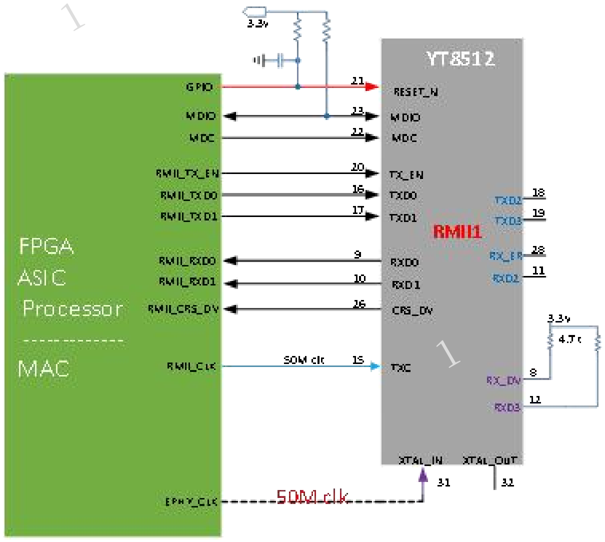
Note:

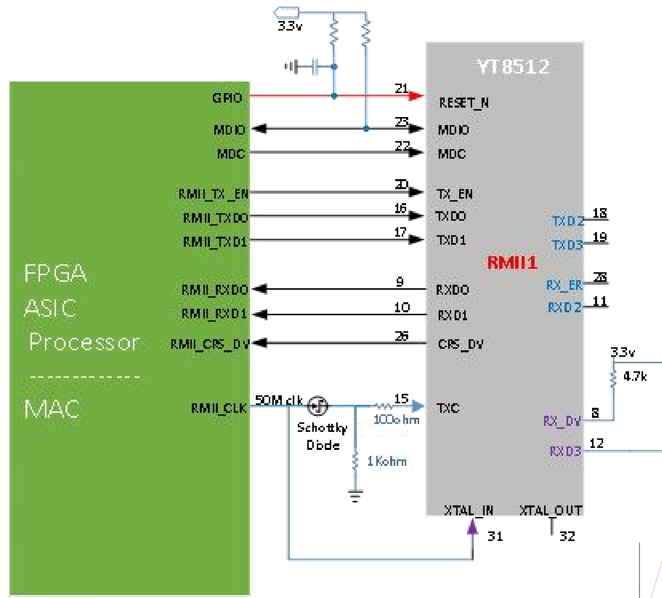
1. The CRS_DV of the PHY in the above figure is connected to the MAC instead of RXDV.

4.8.2.2 RMII1/2 (50MHz clock)

When 50MHz is used as the reference clock, in addition to setting the POS to the correct mode, the PHY needs to be configured with the appropriate software during power-on initialization.

Usually, when 50MHz is used as the reference clock, it is the 50MHz clock provided by the MAC: it can be the TXC of the MAC, or the MAC has a separate 50MHz clock signal output. At this time, PHY is used in RMII1 mode. The following figures take these two cases as examples (Using a 50MHz crystal is similar, please refer to the clock section above), showing the connection between PHY and MAC:





Hardware POS Settings:

RXDV and RXD3 are externally connected to pull-up

When the hardware 50MHz clock is divided into two:

A Schottky diode and two resistors are required for isolation and current limiting. Most Schottky diodes are available, such as SD103AWS.

Software register settings:

Write_ext_reg0x50[6]: 1’b1 //Select 50MHz reference clock

Write_ext_reg0x4000[4]: 1’b1//In RMII mode, the default is CRS_DV, if it needs to be configured as RX_DV, it needs to be set to “1”

Write_mii_reg0x0[15]: 1’b1 //soft reset

Notes:

1. The CRS_DV of the PHY in the above figure is connected to the MAC instead of RXDV.
2. The above software register needs to be set as long as it is 50MHz, no matter it is external input/crystal, or no matter it’s RMII1/RMII2.
3. The voltage, clock, and reset sequence at power-on must meet the requirements mentioned in “4.2 Power-on Sequence”. Because the clock is provided by the CPU, the reset pin of PHY should also be controlled by the CPU to facilitate the adjustment to meet the power-on sequence requirements.

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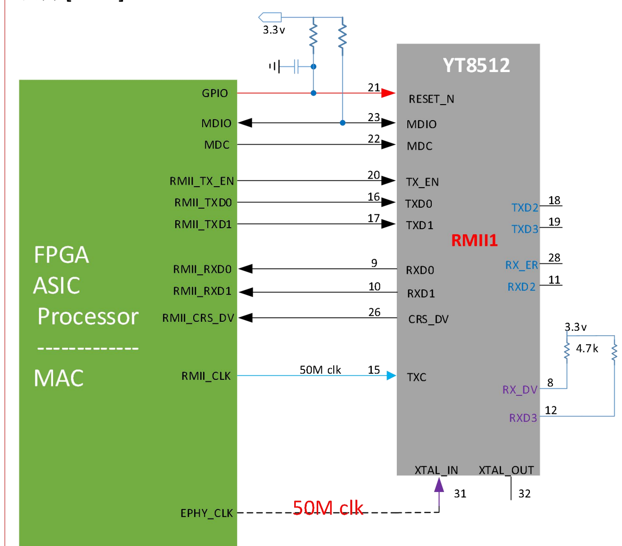

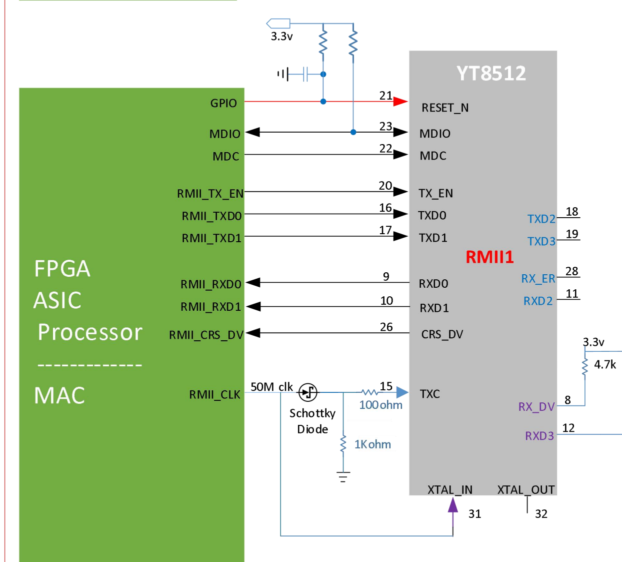

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4. The above-mentioned circuitry for application scenario with external input of 50MHz is inconsistent with the benchmark product, and special attention should be paid to the replacement program.
5. When the clock is externally input, it is recommended to input it from XTAL_IN, because XTAL_IN is in a high-impedance input state and will not affect the input clock amplitude.

4.9 Avoid entering SCAN mode

SCAN mode is a test mode of YT8512. It is used for chip factory testing, and cannot enter this mode during normal use, otherwise the chip will not work normally, and even MDIO will not be accessible. YT8512 makes the chip enter SCAN mode by inputting high level of CRS_DV pin during power-on reset. This pin has a weak pull-down inside the chip and is in an output state. Theoretically, it is low level during power-on reset and the connected counterpart pin should be in input state. If the counterpart pin has its own pull-up or it pulls this pin strongly to high during PHY power-on reset, the YT8512 will enter SCAN mode by mistake, resulting in abnormal function.

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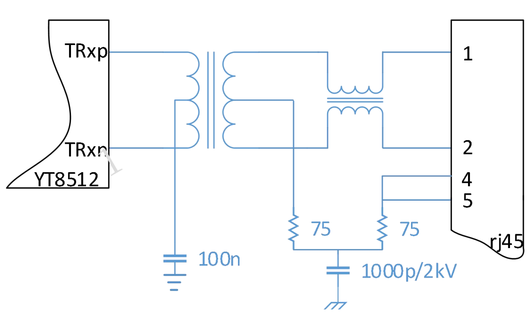
How to avoid entering SCAN mode:

1. Make sure that the other end connected to the YT8512 CRS_DV pin is in input state and there is no pull-up resistor, and then no additional circuit is required.
2. If there is a pull-up resistor at the other end, it is necessary to add an additional pull-down 4.7Kohm resistor to CRS_DV to ensure that it is in a pull-down state.
3. It is measured that CRS_DV should remain low for at least 10us after the power-on hard reset signal is released.

4.10 MDI interface



YT8512 has a built-in port self-rotation function, which can automatically identify straight-through cables or crossover network cables. YT8512 has a built-in 100Ω termination resistor and voltage-type output, so the center tap on the chip side of the transformer only requires a 100n capacitor to ground.

Notes:

1. Only the connection method of one pair of MDI cables is shown in the figure above, and the connection method of the other pair of MDI cables is the same.
2. Since 100M only uses two pairs of cables, the other two pairs of cables on the RJ45 can be connected according to the RJ45 pin 4, 5 in the above figure.

4.11 Delayed MDIO operation requirements after hard reset

After the hard reset signal changes from low to high, the chip needs to complete the internal reset and power-on-strapping work within a period of time, so it is generally required to perform MDIO operations after the hard reset is released for at least 100ms.

4.12 Jitter requirements for external reference clock

Usually, passive crystals can be connected to the XTL_IN and XTL_OUT pins to generate the reference clock for the DUT. For an external clock signal, please refer to the datasheet for the specific connection method. The clock jitter range that requires external input of 25MHz is as follows:

Peak to peak jitter < 200ps;

25kHz~25MHz rms jitter < 3ps;

broadband rms jitter < 9ps

(For other requirements, such as amplitude, duty cycle, etc., please refer to the datasheet)

4.13 Operational Requirements for External Reference Clock Source Switching

In the case of an external reference clock, if there are multiple clock sources and clock source switching is required, then the requirements for the switching process are as follows:

1. during the switching process, the behavior of the clock and the impact on the chip are unpredictable and the hard reset is more thorough than the soft reset, it is recommended from the perspective of system reliability to reset the PHY with a hard reset after switching the clock source.
2. For the sequence of hard reset, please refer to the reset sequence requirements in this application note (that is, the RESET signal should be kept low for at least 10ms after the clock is stabilized, and then pulled high/released).
3. After the hard reset is released, all register settings configured by the previous software will be cleared. MDIO needs to wait for at least 100ms before operation. The software can

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reconfigure the required registers through MDIO (generally the same as the register configuration for power-on initialization).

4.14 Data transmission path delay

When the system implements the precise clock synchronization function, it is necessary to know the time taken for data to transfer from the MDI interface to the MII/REMII/RMII interface, or from MII/REMII/RMII interface to the MDI interface. It includes the absolute value of time, and the jitter range of time (variation). In each mode, its values are as follows:

mode	direction	speed	basic delay(ns)	variation(ns)
mii_mode	mii_tx->utp_tx	100M	232	0
	utp_rx ->mii_rx	100M	204	40
	mii_tx->utp_tx	10M	1170	0
	utp_rx ->mii_rx	10M	7060	400
rmii_mode	rmii_tx->utp_tx	100M	392	40
	utp_rx ->rmii_rx	100M	424	80
	rmii_tx->utp_tx	10M	2770	400
	utp_rx ->rmii_rx	10M	9460	800
remii_mode	remii_tx->utp_tx	100M	472	40
	utp_rx ->remii_rx	100M	424	80
	remii_tx->utp_tx	10M	3570	400
	utp_rx ->remii_rx	10M	9460	800

Note:
The values in the above table have certain errors, and the specific data are subject to the actual measurement results.

5. Register access method

The system relies on MDIO interface communication to read and write the internal registers of YT8512 for configuration management and status acquisition.

5.1 Types of registers

YT PHY has three types of internal registers inside it: MII register (hereinafter abbreviated as mii_reg), extended register (hereinafter abbreviated as ext_reg) and MMD register (hereinafter abbreviated as mmd_reg). All the three registers are addressed separately, and all addresses start from 0x00.

mii_reg: The registers are addressed from 0 to 0x1f, and are accessed in accordance with the standard definition of 802.3 clause 22. The MDC/MDIO protocol is as follows:

Table 22–12—Management frame format

	Management frame fields							
	PRE	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
READ	1...1	01	10	AAAAA	RRRRR	Z0	DDDDDDDDDDDDDDDD	Z
WRITE	1...1	01	01	AAAAA	RRRRR	10	DDDDDDDDDDDDDDDD	Z

ext_reg: Since mii_reg only has 0~0x1f, that is 32 registers. If the product's requirement for the number of registers cannot be met, more registers are extended by accessing two mii_reg 0x1e, 0x1f. The access method is:

Write the ext_reg address to be accessed to mii_reg 0x1e.

Read mii_reg 0x1f to get the value, which is the value in ext_reg. Write the value to mii_reg 0x1f, that is, change the content of ext_reg register to the written value.

For example: read the value of ext_reg0x1000: write 0x1000 to mii_reg0x1e;

read mii_reg0x1f;

write 0x3456 to ext_reg0x1000:

write 0x1000 to mii_reg0x1e;

write 0x3456 to mii_reg0x1f;

mmd_reg: In order to extend more registers to accommodate higher speed Ethernet, MMD registers are created. The MMD registers are accessed in accordance with the standard definition of 802.3 clause 45. MMD registers have two access methods: indirect access and direct access.

Direct access: Its protocol is different from that of MII access. It realizes the reading or writing of the MMD register through two instructions, that is, the address instruction is sent first, and then the Write or Read instruction is sent. The specific protocol is as follows:

Table 45–153—Extensions to management frame format for indirect access

	Management frame fields							
Frame	**PRE**	**ST**	**OP**	**PRTAD**	**DEVAD**	**TA**	**ADDRESS / DATA**	**IDLE**
Address	1...1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1...1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1...1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read-increment-address	1...1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

Indirect access: the MMD registers are accessed through the MII registers (mii_reg 0xd, mii_reg 0xe, see the table below for their specific meanings).

For example:

Read the 0x5 register of MMD 3, and the instruction is:

write mii_reg 0xd 0x3; write mii_reg 0xe 0x5; write mii_reg 0xd 0x4003; read mii_reg 0xe

Write 0x6 to the 0x3c register of MMD 7, and the instruction is:

write mii_reg 0xd 0x7; write mii_reg 0xe 0x3c; write mii_reg 0xd 0x4007; write mii_reg 0xe 0x6

MII 0Dh: MMD access control register

Bit	Symbol	Access	Default	Description
15:14	Function	RW	2'b0	00 = Address 01 = Data, no post increment 10 = Data, post increment on reads and writes 11 = Data, post increment on writes only
13:5	Reserved	RO	9'b0	Always 0
4:0	DEVAD	RW	5'b0	MMD register device address. 00001 = MMD1 00011 = MMD3 00111 = MMD7

MII 0Eh: MMD access data register

Bit	Symbol	Access	Default	Description
15:0	Address data	RW	16'b0	If register 0xD bits [15:14] are 00, this register is used as MMD DEVAD address register. Otherwise, this register is used as MMD DEVAD data register as indicated by its address register.

5.2 Access other types of registers through MII registers

As mentioned above, either ext_reg or mmd_reg can be accessed through mii_reg.

Assume that the functions of read_mii_reg (phy_addr, reg_addr) and write_mii_reg (phy_addr, reg_addr, data) represent reading and writing the address of the specified mii register at the specified phy address. That is:

Reading and writing of Mii registers:

Function name: **read_mii_reg** (phy_addr, reg_addr)

Input parameters: phy_addr, reg_addr

Return value: the read value of the mii reg_addr register

Function name: **write_mii_reg** (phy_addr, reg_addr, data)

Input parameters: phy_addr, reg_addr, data

Return value: None

then:

Reading and writing of Ext registers:

Function name: **read_ext_reg** (phy_addr, reg_addr)

Input parameters: phy_addr, reg_addr

Return value: the read value of the Ext register reg_addr

```
def read_ext_reg (phy_addr, reg_addr):
write_mii_reg (phy_addr, 0x1e, reg_addr)
d = read_mii_reg (phy_addr, 0x1f)
     return d
```

Function name: **write_ext_reg** (phy_addr, reg_addr, data)

Input parameters: phy_addr, reg_addr, data

Return value: None

```
def write_ext_reg(phy_addr, reg_addr, data)
    write_mii_reg(phy_addr, ox1e, reg_addr)
    write_mii_reg(phy_addr, 0x1f, data)
```

Reading and writing of MMD register:

Function name: **read_mmd_reg** (phy_addr, device, reg_addr)

Input parameters: phy_addr, device, reg_addr

Return value: the read value of MMD register

```
def read_mmd_reg(phy_addr, device, reg_addr):
    write_mii_reg(phy_addr, 0x0d, device)
    write_mii_reg(phy_addr, 0x0e,addr)
    write_mii_reg(phy_addr, 0x0d, 0x4000+device)
    d = read_mii_reg(phy_addr, 0x0e)
    return d
```

(Note: for example, MMD3 0x01, where 3 is the device address of the MMD register; 0x01 is the register address of the MMD)

Function name: **write_mmd_reg** (phy_addr, device, reg_addr, data)

Input parameters: phy_addr, device, reg_addr, data

Return value: None

```
def write_mmd_reg(phy_addr, device, reg_addr, data)
    write_mii_reg(phy_addr, 0x0d, device)
    write_mii_reg(phy_addr, 0x0e,reg_addr)
    write_mii_reg(phy_addr, 0x0d, 0x4000+device)
    write_mii_reg(phy_addr, 0x0e, data)
```

5.3 Representation of certain bits of the read-write register

Compared with writing all the bits of a register, most of the functions are only to write some bits of the register, while other bits need to keep the original value unchanged. The way to achieve this is usually to read the original value first, modify some bits of the value, and then write it back to the register. Alternatively, read the value of a register to get only the relevant bits. For convenience, we use the following syntax in this document (take mii_reg0x1, ext_reg0xa as examples):

write_mii_reg0x1[15]: 1'b0:

Write 0 to bit15 of mii register 0x1, a total of 1 bit. The rest of the bits remain unchanged.

read_mii_reg0x1[15]:

Read the value of mii register 0x1 and get bit15, a total of 1 bit.

write_ext_reg0xa[15,14,13]: 3'b010:

Write 0, 1, 0, respectively to bit15, 14, 13 of the extension register 0xa, total 3 bits. The rest of the bits remain unchanged.

read_ext_reg0xa[15,14,13]:

Read the value of extension register 0xa and get bit15, 14, 13.

5.4 Instructions for using phyaddr0 and bdcst_addr

YT8512 has two special functions, phyaddr0 and bdcst_addr, and the users can access the registers of YT8512 conveniently and quickly.

When phyaddr0 is enabled, YT8512 can not only respond to the read/write command with phy address = PHYAD[1:0], but also respond to the read/write command with phy address = 0x0. (For the description of Strapping PHYAD[1:0], please refer to Datasheet.)

Enable phyaddr0 by write_ext_reg0x0[6]:1’b1, and disable phyaddr0 by write_ext_reg0x0[6]: 1’b0. phyaddr0 is enabled by default.

When bdcst_addr is enabled, YT8512 can respond to the write command with phy address = bdcst_addr to achieve the purpose of configuring multiple PHYs on the MDIO bus at one time. Among them, bdcst_addr can be configured to any value from 0x0 to 0x1f.

Use write_ext_reg0x0[5]: 1’b1 to enable bdcst_addr, and write_ext_reg0x0[4:0] to configure the corresponding broadcast address.

Use write_ext_reg0x0[5]: 1’b0 to disable bdcst_addr. By default, En_bdcst_addr is enabled and Bdcst_addr = 0x1f.

It is important to note that:

1. When bdcst_addr is enabled, YT8512 will not respond to the read command with phy address = bdcst_addr, but only respond to the write command;
2. When phyaddr0 is enabled, YT8512 can respond to the read command with phy address = 0x0. However, if there are multiple PHY chips on the MDIO bus, the value read back at this time is unreliable.

EXT 0000h:				
Bit	Symbol	Access	default	Description
15	Reserved	RO	1’b0	Reserved
14	Res_calib_comp_dout	RO	1’b0	Resistor calibration’s compare result.
13:08	Res_calib_config	RO	6’b0	Resistor calibration result
7	En_1588_addr_incr	RW	1’b0	Enable addr increase automatically for 1588 reg access to speed up.
				No use.
6	En_phyaddr0	RW	1’b1	Enable mdio phy address 0 access
5	En_bdcst_addr	RW	1’b1	Enable mdio broadcast address access
4:00	Bdcst_addr	RW	5’b11111	MDIO Broadcast address

6. Register function configuration

YT8512 is a single-port 100M Ethernet PHY. This document mainly introduces some of its usage, including: how to implement some special applications on hardware, how to configure registers for specific functions on software, and frequently asked questions in hardware circuit design.

6.1 Initialization configuration and unsupported functions

Known DUTs may have problems in some applications, which need to be avoided by initializing the configuration. Unless otherwise specified, these configurations may be performed only once after power-on reset, and only need to be reconfigured after power-off or hard reset. The following are the configurations that need to be done during the initialization phase.

6.1.1 100M half-duplex mode fails in MII mode

In MII mode, when the connection status is 100M half-duplex, the TXC/RXC of the MII interface will have no output, resulting in no communication between the MAC and the PHY. To avoid this problem, the following settings need to be done during initialization (or after hard reset):

write_ext_reg0x200a[15]: 1’b0 (clear bit15 of the extension register 0x200a, and keep other bits unchanged).

Notes:

1. There is no such problem in RMII mode, and this initialization configuration is not required.
2. This problem is also described in detail in YT8512 errata notice.

6.1.2 Connecting to 10BT by mistake without plugging in the network cable

If YT8512 is connected to 10BT and the network cable is disconnected, then there is a probability that YT8512 still shows that it is connected to 10BT. To avoid this problem, the following settings need to be made during initialization (or after hard reset):

write_utp_ext_reg0x200a [10]: 1’b0. (Clear bit10 of the extension register 0x200a, and keep the other bits unchanged).

Note:

1. This problem is also described in detail in YT8512 errata notice.

6.2 UTP (electrical port) configuration

6.2.1 10/100BT speed and duplex

After power-on hard reset, phy mii_reg0x2,0x3 can be read to get the PHY ID. By default, YT8512 does not need any configuration, its auto-negotiation capability is enabled, and 100/10BT capability is enabled. The phy will automatically establish a connection with the peer without the intervention of the main CPU (MAC).

How to change the speed and duplex supported by PHY? Including:

1. changing the speed and duplex capability supported by PHY in auto-negotiation mode:

   write_mii_reg0x0[12]: 1’b1 #auto-negotiation is open, this is the default open state write_mii_reg0x4[8:5] #set the corresponding bit to 1 to enable the corresponding speed duplex capability of 100/10BT during local auto-negotiation

write_mii_reg0x0[15]: 1’b1 #soft reset to enable the above configuration

2. changing the speed and duplex capability supported by PHY in forced mode:

   write_mii_reg0x0[12]: 1’b0 #autonegotiation capability is off

   write_mii_reg0x0[6,13] #set the corresponding bit to 1 to enable the corresponding local mandatory 100/10BT speed duplex capability

   write_mii_reg0x0[15]: 1’b1 #soft reset to enable the above configuration

Notes:

1. Generally, it is not recommended to configure into the forced mode as it will make PHY or the other side work in half-duplex mode, while half-duplex, an old transmission mode, will cause packet collision, which will seriously affect the speed or cause packet loss.
2. In the auto-negotiation mode, only the local UTP speed and dual-open capability are turned on or off. In the process of establishing a connection with the other side, the two sides transmit their supported capabilities online, and finally connect according to the highest capability supported by both sides.

How to determine the current connection speed and duplex of the PHY? There are two methods:

Use the standard PHY MII registers to determine whether the current is linked up or not (phy mii_reg0x1[2]), and then use phy mii_reg0x4, 0x5, 0x9 and other standard registers and algorithms to get the current PHY connection speed and duplex.

In order to get the information of current PHY connection, speed and duplex more conveniently, YT PHY puts this information in phy mii_reg0x11 (read-only register), the system only needs to read the 0x11 register, configure the MAC accordingly for normal communication.

The meanings of several bits are as follows (or refer to the chip manual):

Bit 15-Bit14:speed mode, 11-- system reserved; 01--100M; 00--0M

Bit 13: Duplex, 1--Full duplex; 0--half duplex

Bit 10: Link status, 1--link up; 0--link down

Note: It is necessary to judge that bit 10 is 1 first, indicating that the connection is established, and then judge bit 15, 14, 13 to obtain the speed and duplex.

After configuring the MAC, confirm that the PHY is connected by polling the above-mentioned registers, and then the upper layer can carry out network communication. One point to note in this process is that after polling the local PHY connection, the peer PHY connection may not be established yet, so the upper layer needs to delay for a while (about 0.5 seconds later) before sending the packet to ensure that the peer connection is established at this time, so as not to cause packet loss.

6.2.2 power down setting

When UTP connection is not required, UTP can be set to power down mode to achieve power saving function:

Enter utp power down: phy mii 0.11 is set to 1

Exit utp power down: phy mii 0.11 is set to 0, phy mii 0.15 set to 1 for software reset operation.

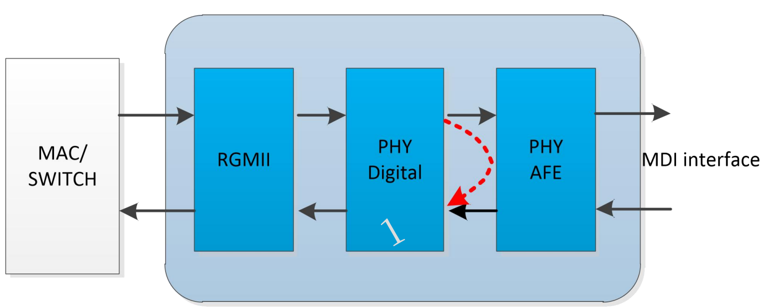
6.2.3 Loopback mode

1. In the case of internal loopback, the analog circuit of utp is bypassed, and the data sent by mac is looped back directly from the digital circuit of utp phy, which can also be called digital loopback, as shown in the figure below. The configuration method is as follows:

   10M: utp phy mii register 0x0 = 0x4100

   100M: utp phy mii register 0x0 = 0x6100

Note: The general register configuration will not be cleared by soft reset, but the two bits of the internal loopback and power down in UTP mii register 0x0 will be cleared by soft reset, so do not do soft reset after enabling these two functions.



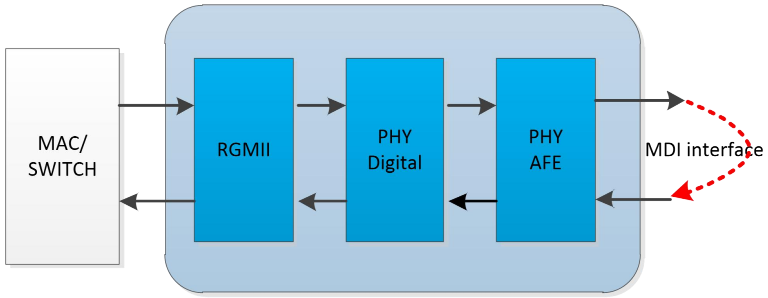
2. In the External Loopback mode, the signal sent by the AFE is returned by the loopback line directly to the receiving circuit of the AFE. MAC ensures the correctness of MII/REMII/RMII interface, PHY digital and analog functions by comparing MII/REMII/RMII transmission data and receiving data. The loopback cable connects pair1&2 to pair3&6 of the network cable, and

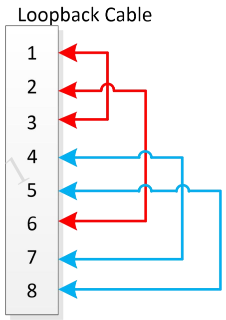
connects pair4&5 to pair7&8, as shown in the figure below. The configuration method is as follows:

10M: first clear phy ext 0x2027 bit15 to 0 (turn off sleep mode); utp phy ext register 0x4000.12= 1, mii register 0x0 = 0x8100;

100M: first clear phy ext 0x2027 bit15 to 0 (turn off sleep mode); utp phy ext register 0x4000.12 = 1, mii register 0x0 = 0xa100

Note: The last mii register 0x0 of the above configuration realizes soft reset as bit15 is set to 1, which makes the previous configuration take effect and initiates the external loopback mechanism. At this time, it should be ensured that the external loopback line has been plugged in, otherwise, it will cause the external loopback to be unable to connect, which then requires another soft reset.





3. In Remote Loopback mode, the data received by MII/REMII/RMII RX is directly injected into MII/REMII/RMII TX. The remote MAC ensures that, by comparing the data sent and received, it and the link partner work well, as shown in the figure below. The configuration method is as follows:

10M: utp ext register 0x4000.11 = 1

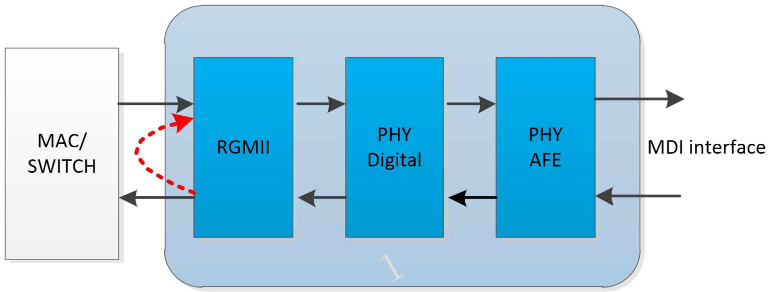
utp phy mii register 0x4 = 0xc41,

utp phy mii register 0x0 = 0x9140,

100M: utp extregister 0x4000.11 = 1

utp phy mii register 0x4 = 0xd01,

utp phy mii register 0x0 = 0x9140



6.2.4 Smart downgrade (auto-downgrade) configuration

For Ethernet PHY, when the network cable is plugged in, by default, it is to try to connect from the highest rate supported by both sides. Sometimes due to external interference or long connection line or poor quality, the high-speed connection cannot be made. When repeated connection attempts exceed a certain number of times, the PHY can choose to automatically downgrade the speed to complete the connection. Smart speed, or auto-downgrade, refers to the function of PHY to automatically reduce the connection speed.

For example, if both sides initiate auto-negotiation according to the capacity of 100BT, but the network cable exceeds the maximum connection distance (for example, to 200 meters), and both sides fail to connect to 100BT after 5 failed attempts (each link up lasts less than 2 seconds), then the side with auto-downgrade function will initiate auto-negotiation at 10BT and try a connection at 10BT.

Notes:

1. If the fast plugging and unplugging of the port leads to continuous link up/down in a short period of time, the smart downgrade mechanism will also be triggered.
2. This function only occurs during the connection establishment time. If the connection is made and maintained, the PHY will not automatically downgrade due to packet loss.

The above is the default behavior of YT8512. If this function is not needed, it needs to be disabled via the register. That is, clear the register phy mii_reg0x14 bit5 to 0, and then do a software reset (set phy mii_reg0x0 bit15 to 1) to make it effective. The description is as follows:

Phy MII 14h: Speed Auto Downgrade Control Register				
Bit	Symbol	Access	Default	Description
15:12	Reserved	RO	0x0	Reserved
11	Reserved	RW	0x1	Reserved

10	Reserved	RW	0x0	Reserved
9	Reserved	RW	0x0	Reserved
8	Reserved	RW	0x0	Reserved
7:6	Reserved	RO	0x0	Reserved
5	En_speed_downgrade	RW	0x1	When this bit is set to 1, the PHY enables smart-speed function. Writing this bit requires a software reset to update. This bit will be set to 1'b0 in UTP_TO_FIBER_FORCE and UTP_TO_FIBER_AUTO mode; else set to 1'b1, only take effect after software reset
4:2	Reserved	RW	0x3	Reserved
1	Reserved	RW	0x0	Reserved
0	Reserved	RO	0x0	Reserved

Note: To disable smart speed, it is necessary to disable the PHY at both ends of the network cable. Otherwise, automatic downgrade on either side will cause the connection to slow down.

6.2.5 Sleep (auto-sleep) configuration

After UTP is in the unconnected state for a certain period of time (about 40 seconds), the PHY will automatically enter the SLEEP state. In the sleep state, the PHY will automatically turn off some internal circuits to save power. After entering sleep state, the PHY will periodically send Pulse signals and turn on the signal detection function. Once the signal sent by the other side is received and exceeds the threshold of signal detection, the PHY will immediately open the relevant circuit and enter the normal working state.

The Sleep function is enabled by default, and this function will not affect the normal connection of the PHY. There will be no side effects if the sleep function is turned off.

Turn off this function: write_ext_reg0x2027[15]: 1’b0.

Phy EXT 2027h: Sleep Control1				
Bit	Symbol	Access	default	Description
15	En_sleep_sw	RW	0x1	1 = enable sleep mode: PHY will enter sleep mode and close AFE after unplug cable for a timer;
14	Pllon_in_slp	RO	0x0	1 = keep PLL on in sleep mode;
13	Slp_pulse_sw	RW	0x1	when PHY enter sleep,
12	En_upd_afe_sbs	RW	0x0	When AFE control is changed, no matter it’s triggered by sleep control logic or normal work mode change,
11:6	Reserved	RO	0x0	Reserved

5	Sleeping	RO	0x0	1 = PHY is slept;
4	Gate_25m	RO	0x0	Not used.
3:0	Slp_state	RO	0x0	FSM state of internal sleep control logic.

6.2.6 **Packet generator and sending/receiving packet statistics**

Packet generator; packet checker.

Descriptions of application of packet checker:

1. open packet checker: write ext reg 0x40A0.[15:14]=2'b10
2. disable packet checker: write ext reg 0x40A0.[15:14]=2'b01
3. view checker's statistical results:
4. ext reg 0x40A3~0x40A8, representing the number of correct packets received from MDI, and the register is read clear.
5. ext reg 0x40A9~0x40AC, representing the number of packets received from MDI with crc error, and the register is read clear.
6. ext reg 0x40AD~0x40B2, representing the number of correct packets received from MII/RMII, and the register is read clear.
7. ext reg 0X40B3~0X40b6, representing the number of package received from MII/RMII with crc error, and the register is read clear.

Descriptions of application of packet generator:

1. set the length of the sending packet of the packet generator: write ext reg 0x40A1[15:0]
2. set the number of sending packets: write ext reg 0x40A2[15:0], the value of 0 means packets are sent all the time.
3. set the IPG length: write ext reg 40A0[7:4], which can be left unset when not in use.
4. open the packet generator.
5. write ext reg 0x40A0[15:13] = 3'b100 to configure packet checker and packet generator.
6. write ext reg 0x40A0[12] = 1'b1 to turn on the packet generator and start sending packets.
7. polling ext reg 0x40A0[12], when it is 1'b0, it means that the set number of packets has been sent. When the number of packets is set to 0, write ext reg 0x40A0[12]:1'b0, which means stop sending packets.
8. turn off the packet generator:

9. write ext reg 0x40A0[14]: 1’b1 to turn off the clock of packet generator
10. write ext reg 0x40A0[15, 13, 12]: 3’b010 to turn off packet generator and checker

6.2.7 SNR reading (current connection quality indication)

The bit [14:0] value of the phy ext 0x205A of YT8512 represents the SNR of a pair of MDI lines.

The reading steps are as follows:

Step 1, set bit15 of phy ext 0x2059 to 1

Step 2, read bits [14:0] of phy ext 0x205A respectively

Write it down as MSE(i), i=0,1,2,3

Read each value at least 100 times and take the flat value to get a stable MSE on each pair of MDI lines

Step 4, take the above stable MSE values and calculate the SNR according to the following formula:

100M: 10*log10(32768/mse)

To achieve a stable connection, the SNR value should be greater than 19.13dB (100M corresponds to MSE of about 400);

6.2.8 Template (electrical port index) configuration

The Template test instructions of YT8512 are as follows:

Initial configuration:

None

100BT:

Write_ext_reg0x2027: 0x2026

Write_mii_reg_0x10: 0x2

Write_mii_reg_0x0: 0xa100

10BTe:

Write_ext_reg0x2027: 0x2026

Write_mii_reg_0x10: 0x2

Write_mii_reg0x0: 0x8100

Write_ext_reg0x200a: 0xCA09 #packet with all ones, 10MHz sine wave, for harmonic test

Write_ext_reg0x200a: 0xCA0A #pseudo random, for TP_idle/Jitter/Different voltage test

Write_ext_reg0x200a: 0xCA0B #normal link pulse only

Write_ext_reg0x200a: 0xCA0C #5MHz sine wave

Write_ext_reg0x200a: 0xCA0D #Normal mode

The above settings can pass the template test. Considering different board-level designs, PHY also has rich internal configurations for user-friendly adjustments. Some of the commonly used adjustments are:

A. adjust +/-Vout Differential Output Voltage:

100BT, adjustable ext_reg0x2057, where coarse adjustment is made for bit [14:12], and fine adjustment for bit [11:8]

10BTe, adjustable ext_reg0x2057, where coarse adjustment is made for bit [6:4], and fine adjustment for bit [3:0]

When the above value increases, the output amplitude will increase, and the amplitude will increase by about 25% for each increase of 1 in coarse adjustment, and about 1/16 for each increase of 1 in fine adjustment.

B. adjust rise/fall time:

ext_reg 0x1[9] set to 1

ext_reg 0x14 set to 0x1fe

ext_reg 0x15 set to 0x1fe

6.3 MII/RMII Configuration

6.3.1 Setting of driving capability

MII/RMII driving capability of UTP is reflected in rise/fall time. Large driving capacity means that it can carry a larger load, but it may also introduce EMI problems. The default driving ability of UTP is a trade-off between the two settings and is not the strongest setting. If it is necessary to strengthen or weaken the driving ability, it can be done by configuring the extension register as follows:

Write_ext_reg0x4001[5:4]: the default is 2’b10. 2’b11 is the strongest and 2’b00 is the weakest.

6.4 LED configuration

删除[hs-lll]: **LED light**

YT8512 supports two LEDs: LED0 and LED1. Their behaviors (flashing, on, off) are determined by the configuration of one of their corresponding registers.

删除[hs-lll]: LED light

The register corresponding to LED0 is ext Reg0x40C0; (power-on default value is 0x0311)

The register corresponding to LED1 is ext Reg0x40C3; (power-on default value is 0x0320)

Assign different values to these two registers, then the LED will display the corresponding status. Typical configurations are as follows:

exReg 0x40C 0/0X 40C 3 Configuration value	Behavior
0x0311	The LED is on when the link is at 10M; The LED flashes when the link is at 10M and packet is sent or received
0x0320	The LED is on when the link is at 100M; The LED flashes when the link is at 100M and packet is sent or received
0x30	LED is on when link is up or packet is sent or received; LED is off when link is down
0x1300	LED flashes when packet is sent or received; LED is off when packet is not sent or received

6.5 Interrupt configuration

Pin 11 (INT_N/RXD2) of YT8512 is a multiplexed pin for interrupt and RXD2. When the chip works in RMII mode, pin 11 is used as INT_N; when the chip works in mii and remii modes, pin 11is used as RXD2.

When used as an interrupt, there are several instructions:

1. The interrupt pin is active low.
2. configure the UTP mii reg0x12 register to select which event in the general interrupt can trigger the interrupt signal.
3. when the corresponding interrupt event occurs, the signal of the interrupt pin becomes low, and the CPU reads the interrupt register (mii reg0x13) to check the specific event that currently

triggers the interrupt. This register is automatically cleared to zero when read, and the interrupt signal automatically goes high.

The relevant registers are:

PHY MII 12h: Interrupt Mask Register				
Bit	Symbol	Access	Default	Description
15	Auto-Negotiation Error INT mask	RW	0x0	1 = Interrupt enable
14	Speed Changed INT mask	RW	0x0	1 = Interrupt enable
13	Duplex changed INT mask	RW	0x0	1 = Interrupt enable
12	Page Received INT mask	RW	0x0	1 = Interrupt enable
11	Link Failed INT mask	RW	0x0	1 = Interrupt enable
10	Link Succeed INT mask	RW	0x0	1 = Interrupt enable
9:7	reserved	RW	0x0	No used.
6	WOL INT mask	RW	0x0	1 = Interrupt enable
5	Wirespeed downgraded INT mask	RW	0x0	1 = Interrupt enable
4:2	Reserved	RW	0x0	No used.
1	Polarity changed INT mask	RW	0x0	1 = Interrupt enable
0	Jabber Happened INT mask	RW	0x0	1 = Interrupt enable
PHY MII 13h: Interrupt Mask Register				
Bit	Symbol	Access	Default	Description
15	Auto-Negotiation Error INT	RO	0x0	Error can take place when any of the following
14	Speed Changed INT	RO	0x0	1 = Speed changed
13	Duplex changed INT	RO	0x0	1 = duplex changed
12	Page Received INT	RO	0x0	1 = Page received
11	Link Failed INT	RO	0x0	1 = UTP link down takes place
10	Link Succeed INT	RO	0x0	1 = UTP link up takes place
9:7	Reserved	RO	0x0	No used
6	WOL INT	RO	0x0	1 = UTP received WOL magic frame.
5	Wirespeed downgraded INT	RO	0x0	1 = speed downgraded.
4:2	Reserved	RO	0x0	No used.
1	Polarity changed INT	RO	0x0	1 = UTP revered MDI polarity
0	Jabber Happened INT	RO	0x0	1 = 10BaseT TX jabber happened

Take Link up as an example to generate interrupt events:

Write_mii_reg0x12: 0x400 #bit10 set to 1

After completing the above settings, it is necessary to verify whether this setting can correctly trigger the interrupt or not, the method is as follows:

1. perform a soft reset on YT8512 to disconnect the existing connection:

   Write_mii_reg0x0: 0x9140 #bit15 set to 1

2. wait a few seconds, or poll UTP status register until the link is up.
3. measure that the INT_N pin signal goes low
4. read the interrupt status register to get the triggered interrupt event

   Read_mii_reg0x13 # get the value of bit10 to be 1

   Read_mii_reg0x13 #the obtained value should be all 0
5. measure that the INT_N pin signal goes high

6.6 Other maintenance and test registers

6.6.1 User self-configuration registers

DUT registers are generally only cleared after power-off or hard reset. If the DUT encounters an exception during operation, and the customer wants to confirm whether a power-off or hard reset has occurred when the exception occurs, a designated register is required. Write the value in the initialization, and read this value when an exception occurs. This register is ext_reg0x40B9 and its default value is 0x0.

6.6.2 PLL lock indication

The PLL inside the DUT will lose lock when the clock signal has a large frequency offset, resulting in abnormal internal operation. The steps to check its instructions are as follows:

1. ext_reg0x61[6] is set to 1, which is enabled by default and can be left unconfigured
2. read ext_reg0Xa0[1:0], if these two bits are not 0 at the same time, the PLL is unlocked.

Remark:

Generally, for the input clk, the PLL unlock will not occur until the offset is above 10MHz.

EXT 61h:				
Bit	Symbol	Access	default	Description
6	en_vco_check	RW	0x0	enable or disable VCO_fast or VCO_slow check circuit

EXT A0h:				
Bit	Symbol	Access	default	Description
1	Pll_vco_h	RO	0x0	PLL fast indicator
0	Pll_vco_l	RO	0x0	PLL slow indicator

6.6.3 Clock drift

The frequency offset between DUT and LP is calculated as follows:

1. set ext reg0x2058[14] to '1';
2. set ext reg0x2059[15] to '1';
3. confirm that the DUT and Link partner are in link up status, and the link is at 100BT;
4. read the value of ext reg0x2068, and take the complement of the read value *0.95 as the frequency difference between DUP and Link partner, in ppm

EXT 2068h: PHY Debug, Integrator				
Bit	Symbol	Access	default	Description
15:00	Integrator	RO	16’h0	Channel 0’s integrator. Frquence different between lp and dut. PPM = 0.95 * integrator; integrator is 2's complement value
				Valid when EXT 2058h bit14 en_snap_shot is set, or EXT 2059h bit15 prob_auto is set. Corresponding to these two setting, it is the:
				Integrator at the time snap shot condition is met, or
				Integrator at the time this EXT register is being read.